COMPUTER NETWORKS

CSE 321

How to get the target data (e.g webpage) from servers?

Motivation

We want a standardized and simple protocol to facilitate sharing of information over the Internet.

Locating Resources

We locate resources (webpage, image, file etc.) using URL (Uniform Resource Locator) over the Internet

www.vuawebsite.com/vuapage

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Host name Path

HTTP

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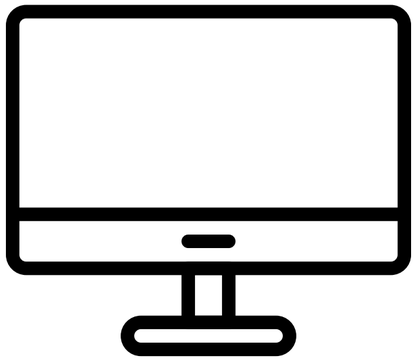
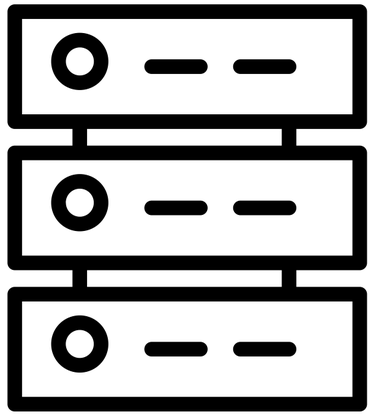


Client



Server

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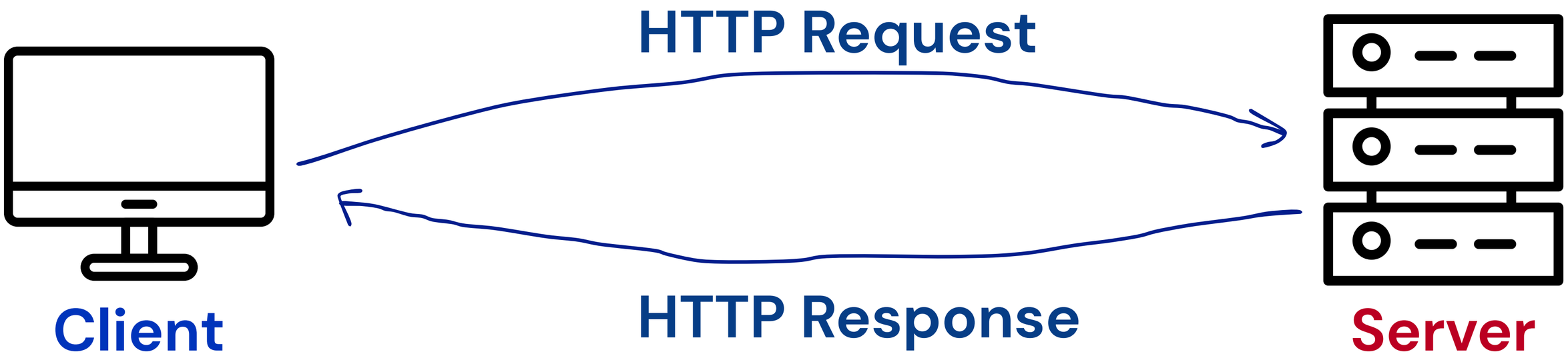


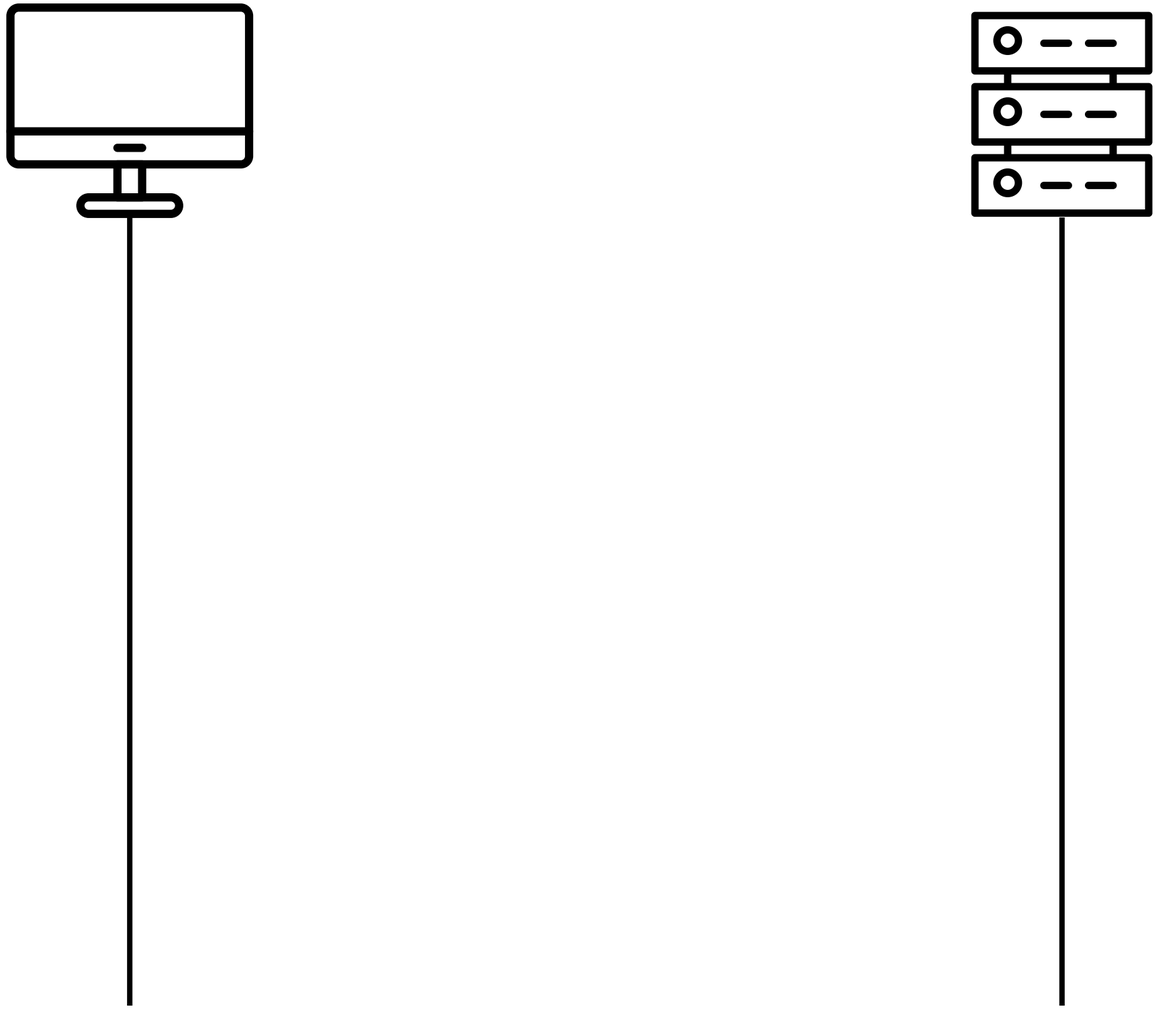
HTTP/0.9

- Only one request method
- Only fetching HTML documents
- No header, metadata, status code

HTTP/1.0

- Uses TCP on transport layer
- Is stateless - maintains no information of past requests (protocol maintaining states are complex)
- Is non-persistent - each request needs a TCP connection

HTTP/1.0



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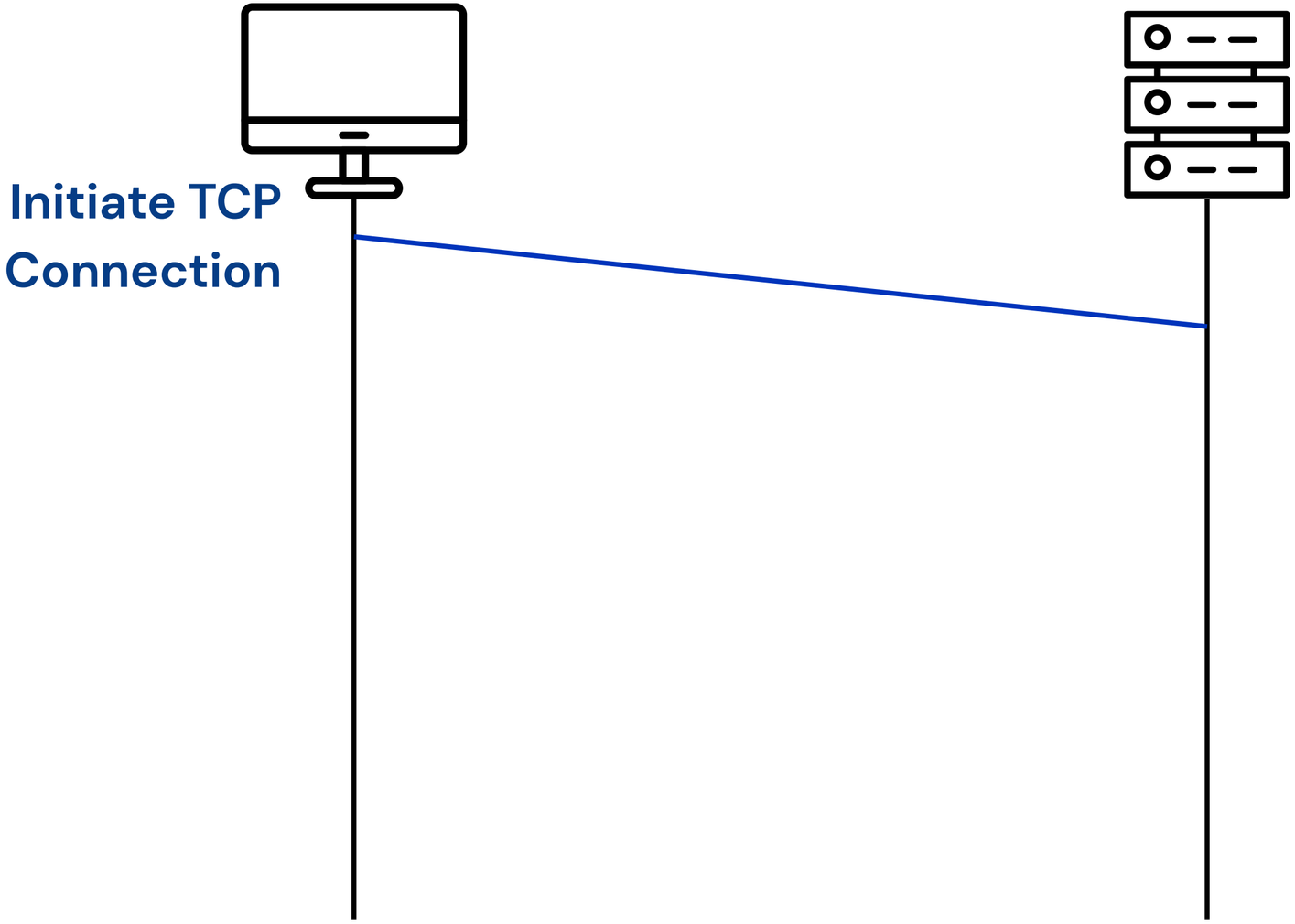

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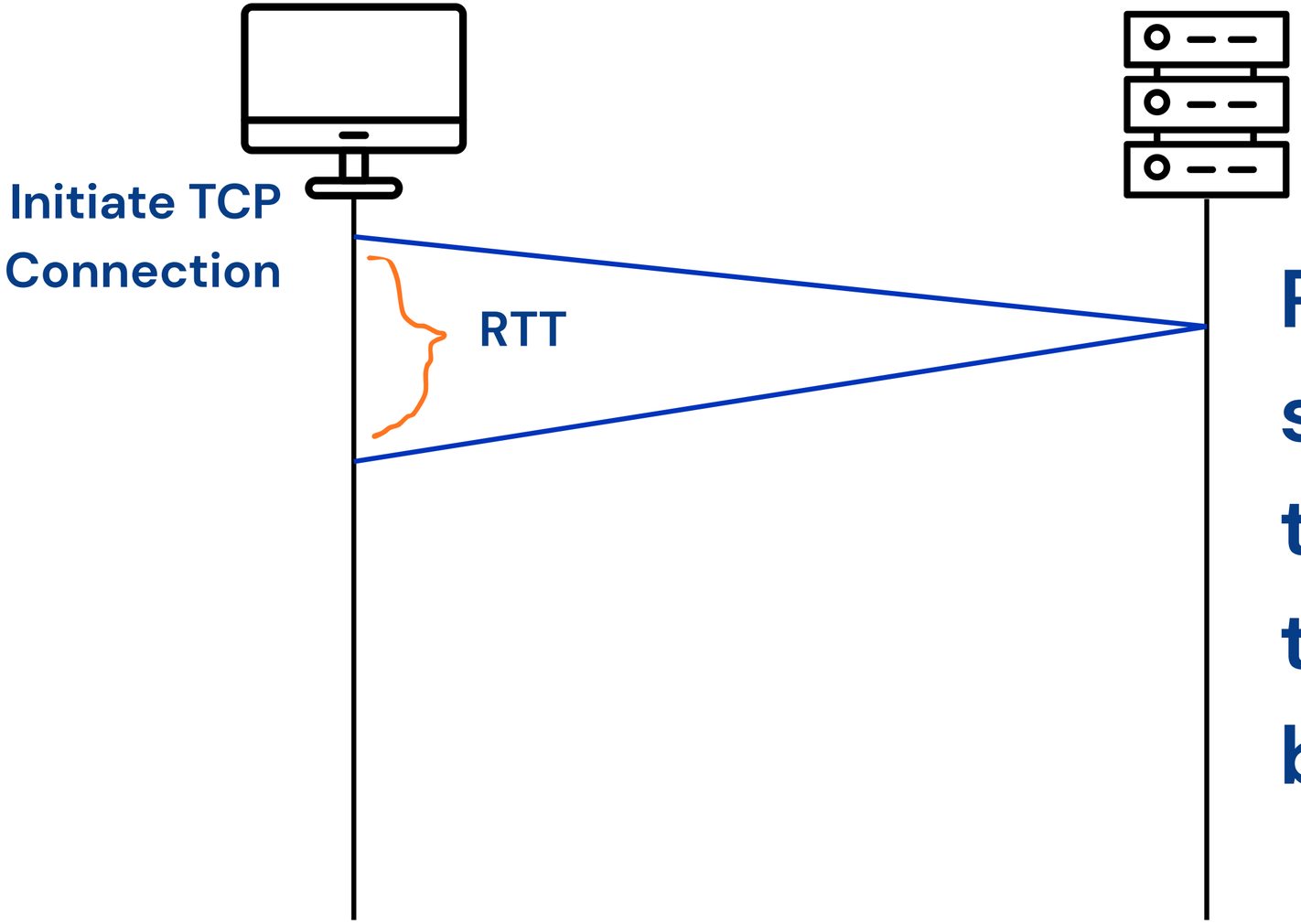
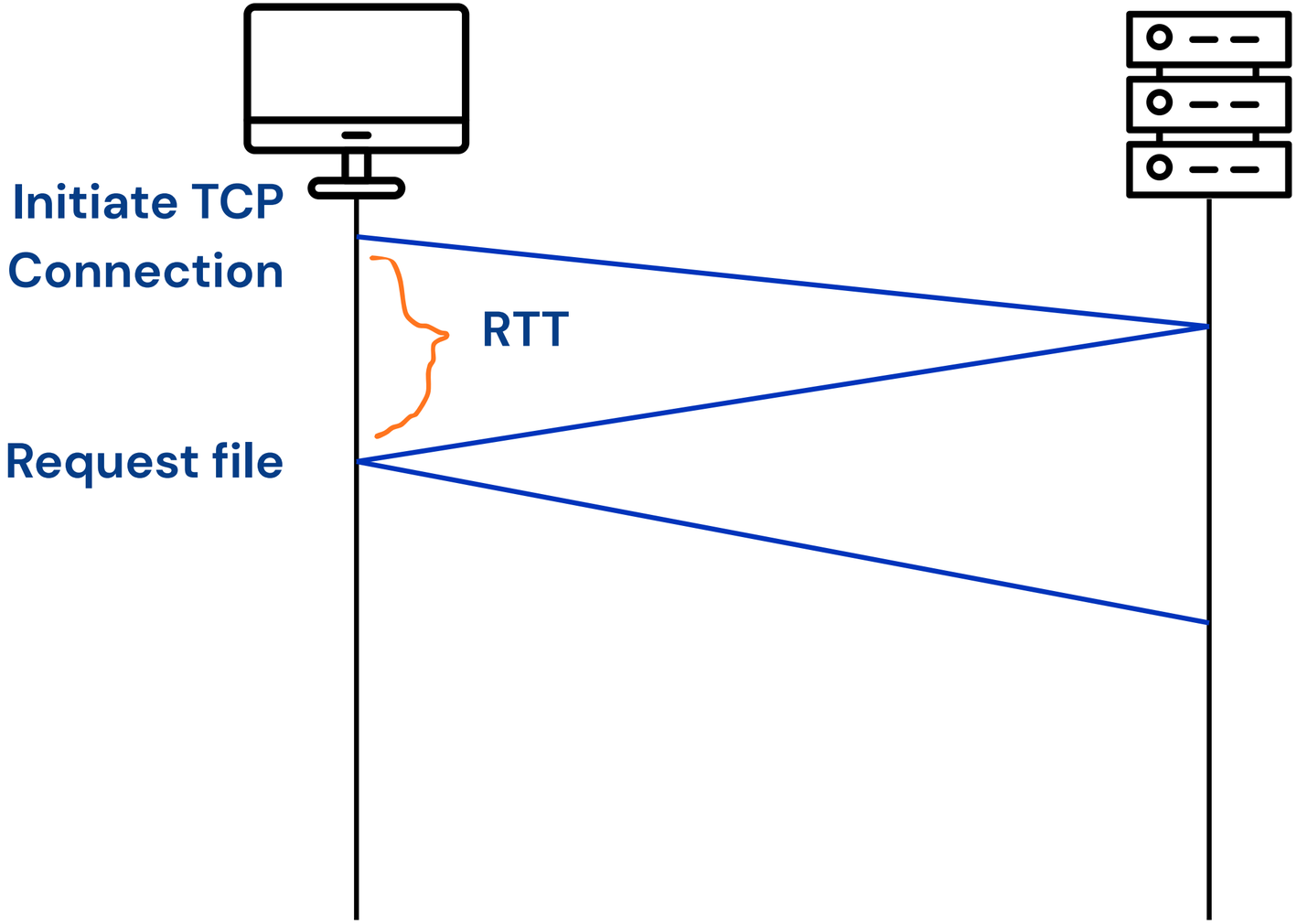


RTT: Time for a small packet to travel from client to server and back

HTTP/1.0



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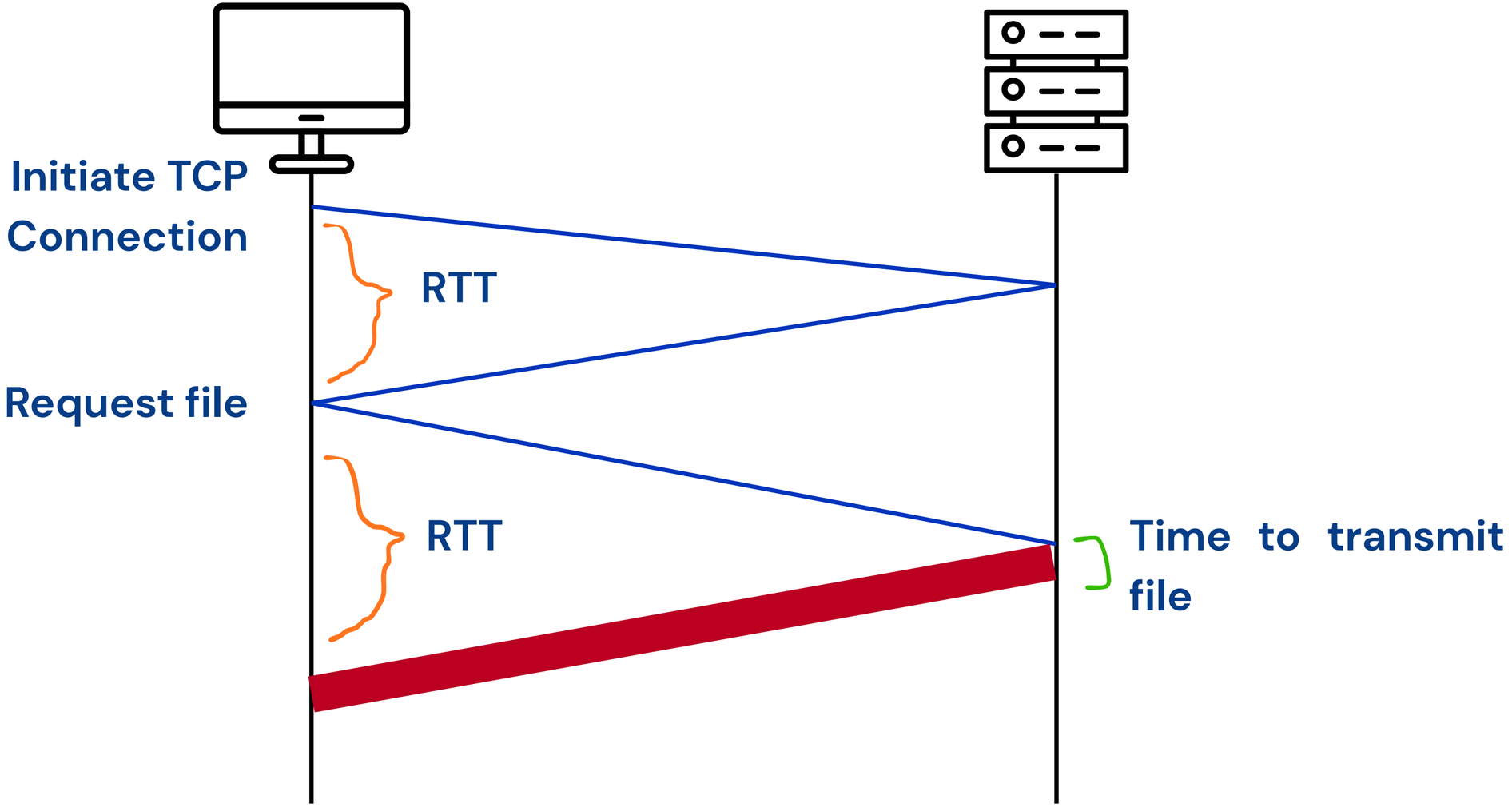
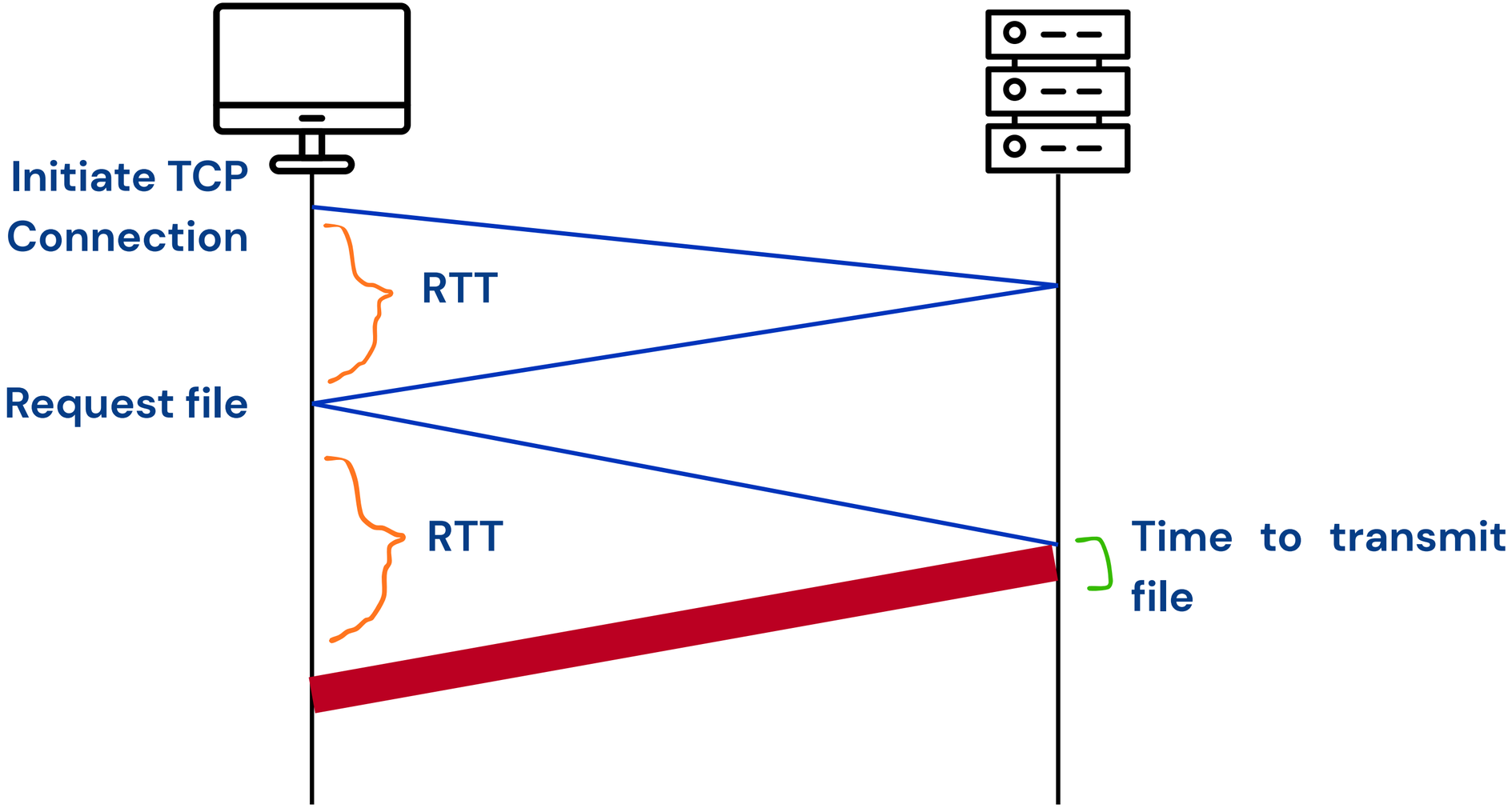
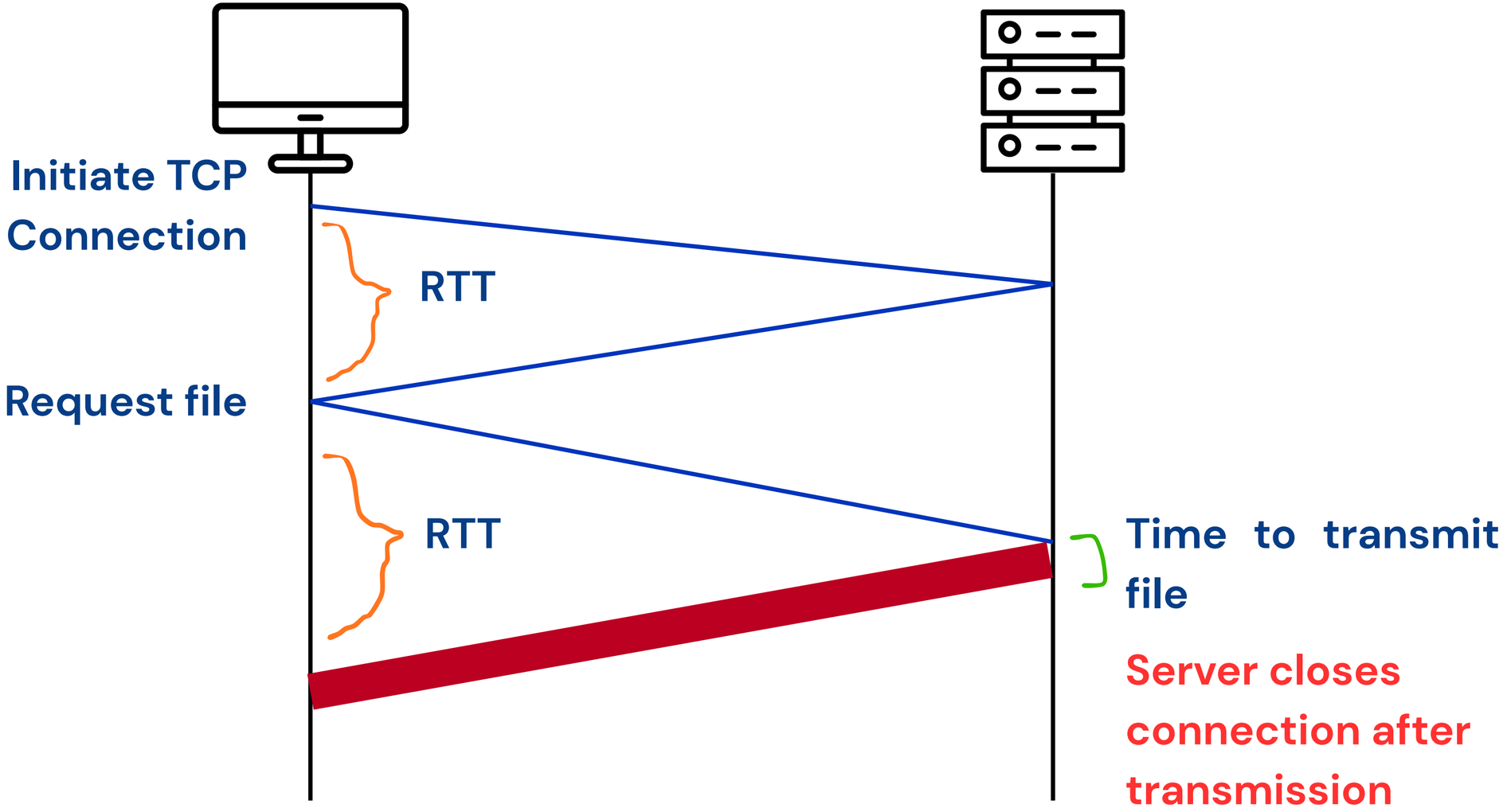

Initiate TCP Connection
RTT
Request file
RTT
Time to transmit file

HTTP/1.0



Response Time = 2RTT + File transmission time

HTTP/1.0



Response Time = 2RTT + File transmission time

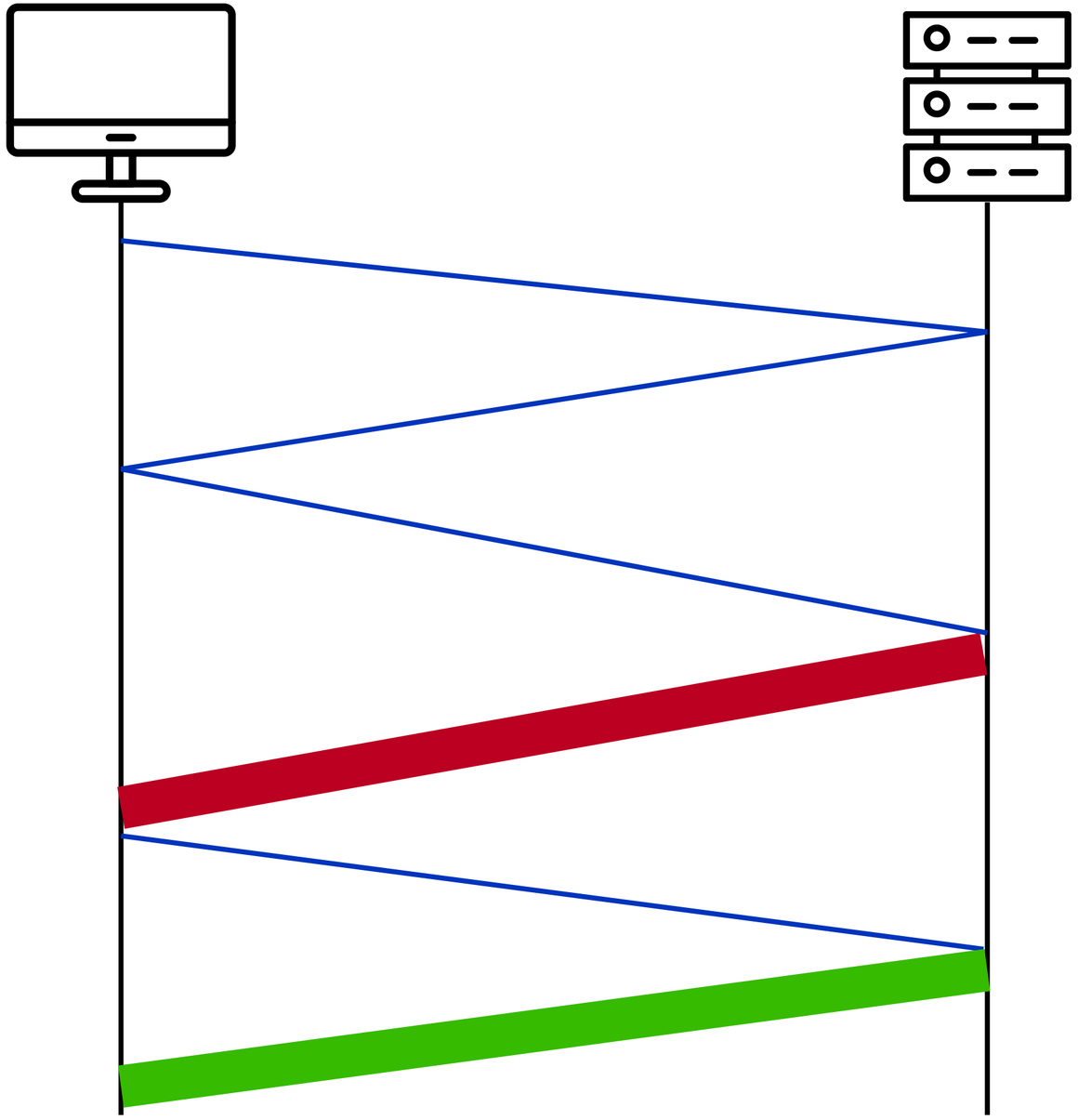
Issues

- 2RTT per object
- OS overheads for each TCP connection

HTTP/1.1 (Persistent HTTP)

- Server leaves connection open after sending response
- Subsequent messages will be sent over the open connection

HTTP/1.1 (Persistent HTTP)



HTTP/1.1 (Pipelining)

Also introduced multiple, pipelined requests. Server responds in FCFS order.

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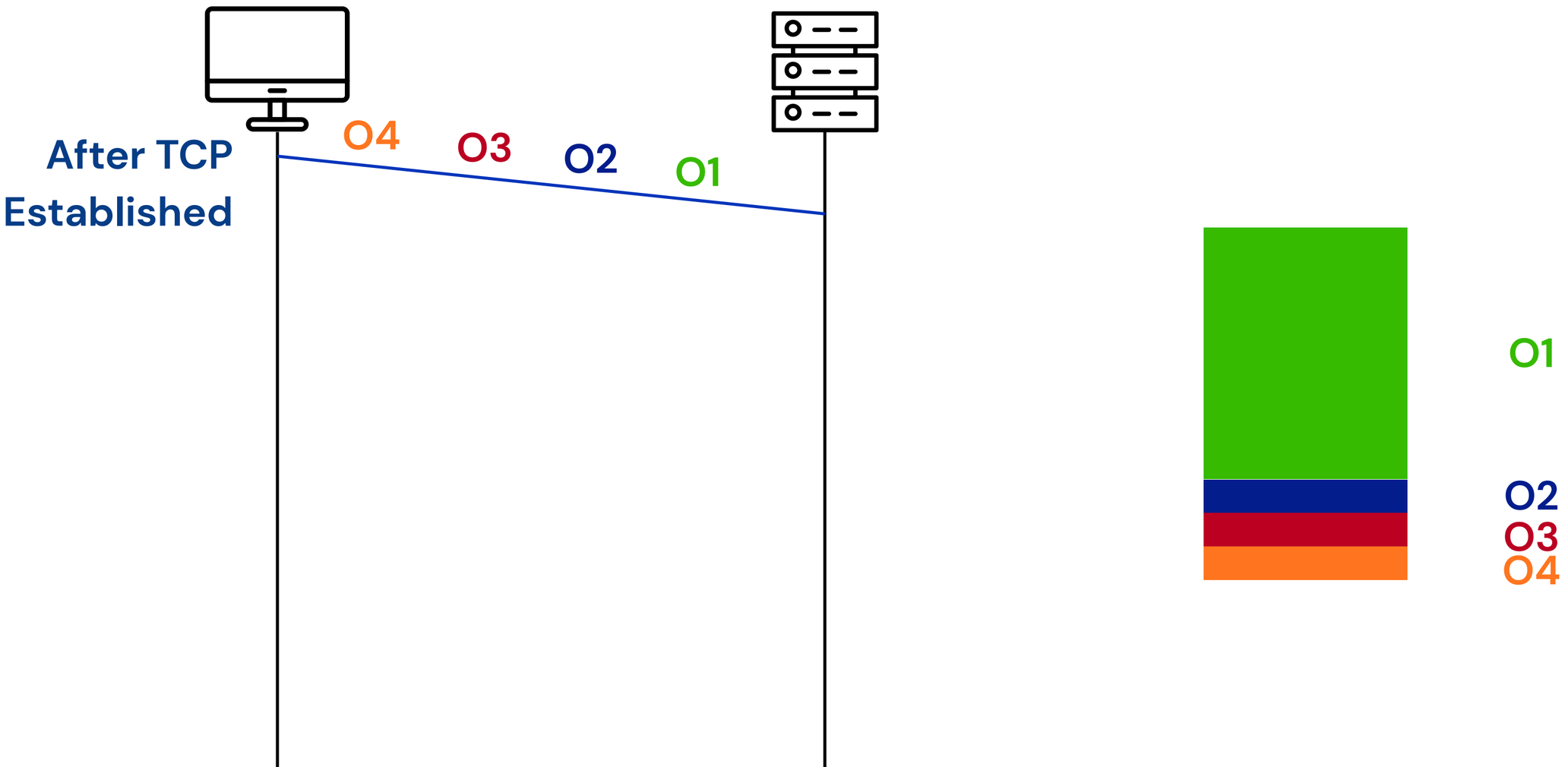

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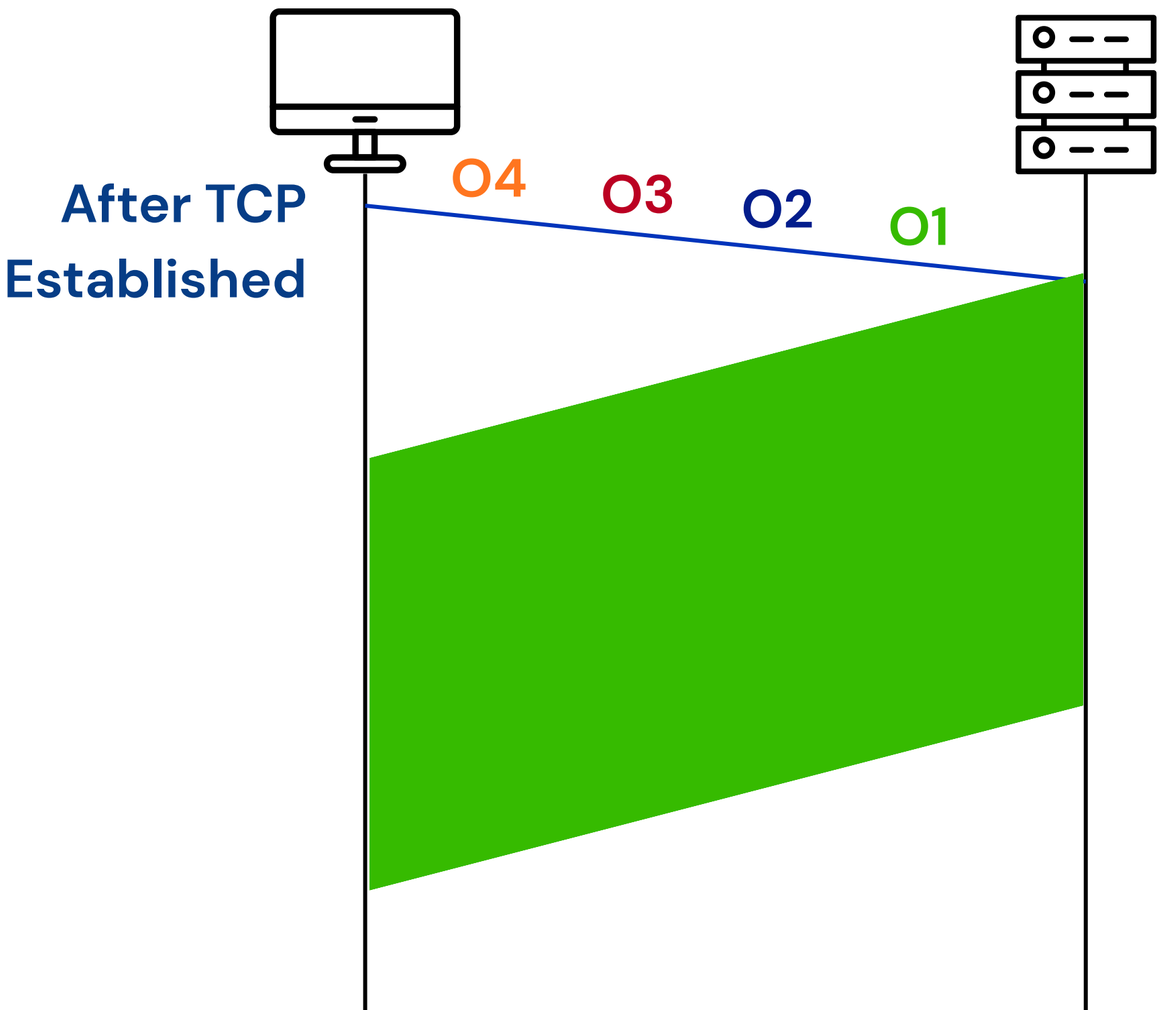

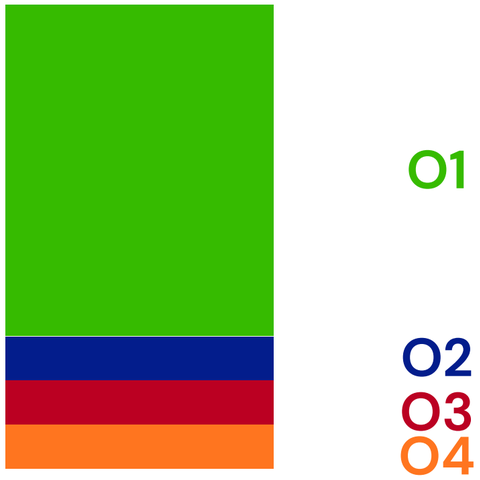

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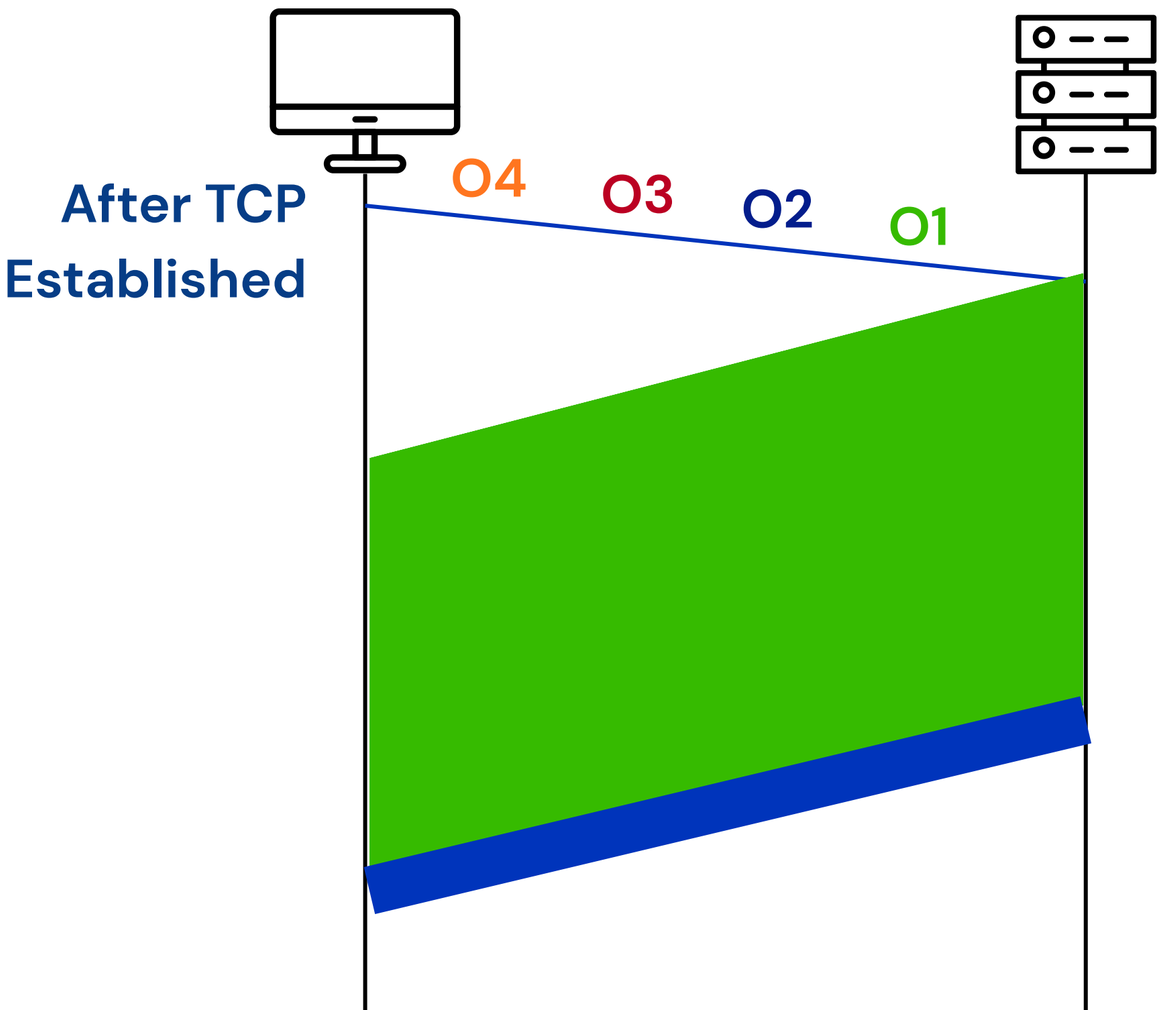

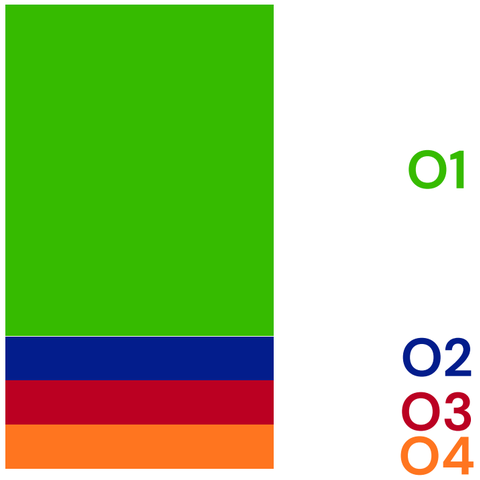

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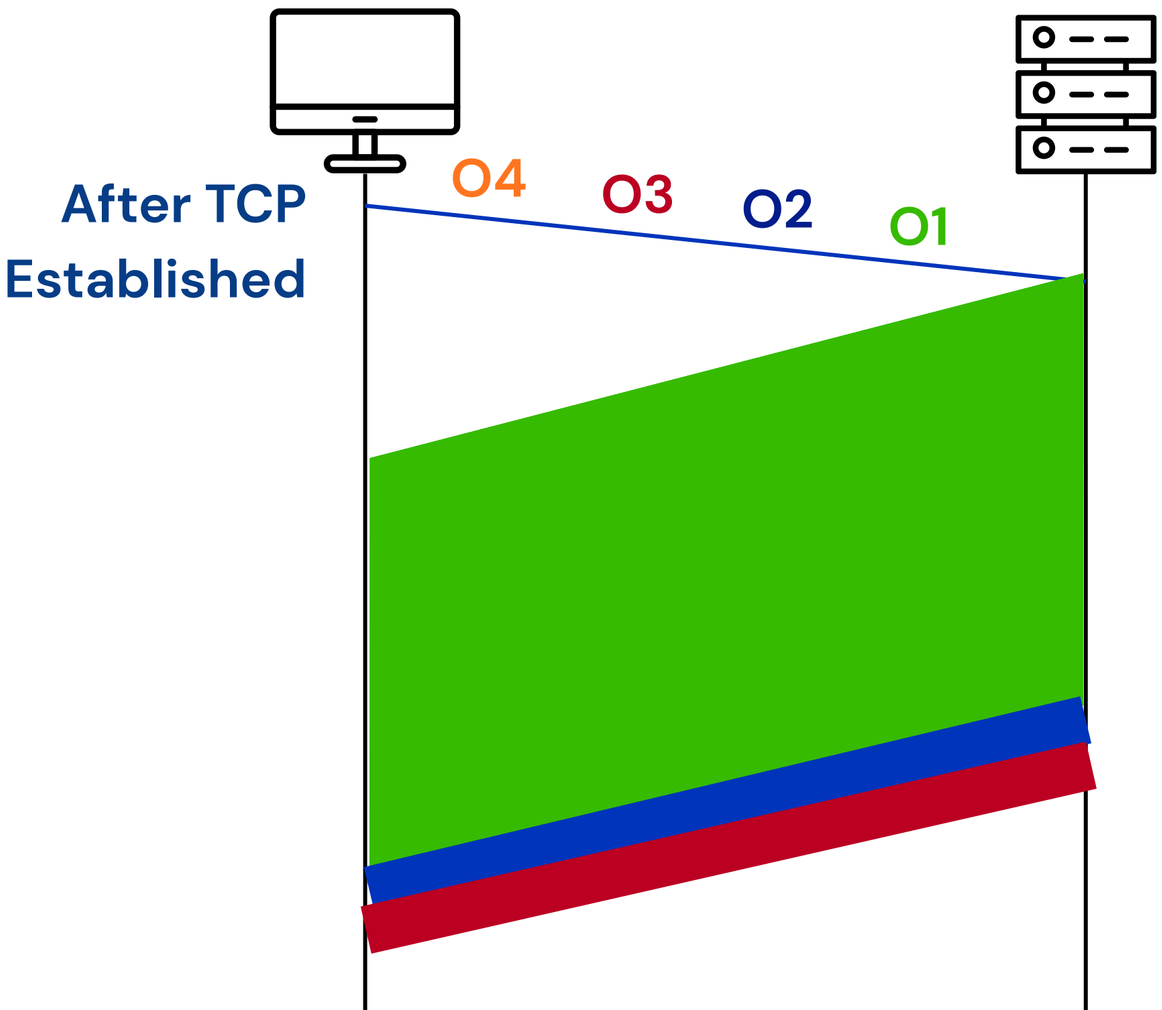

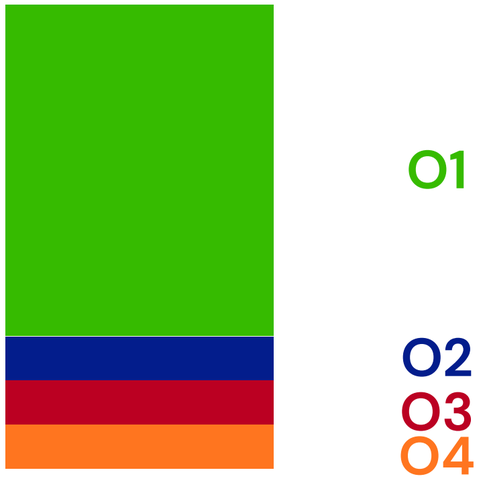

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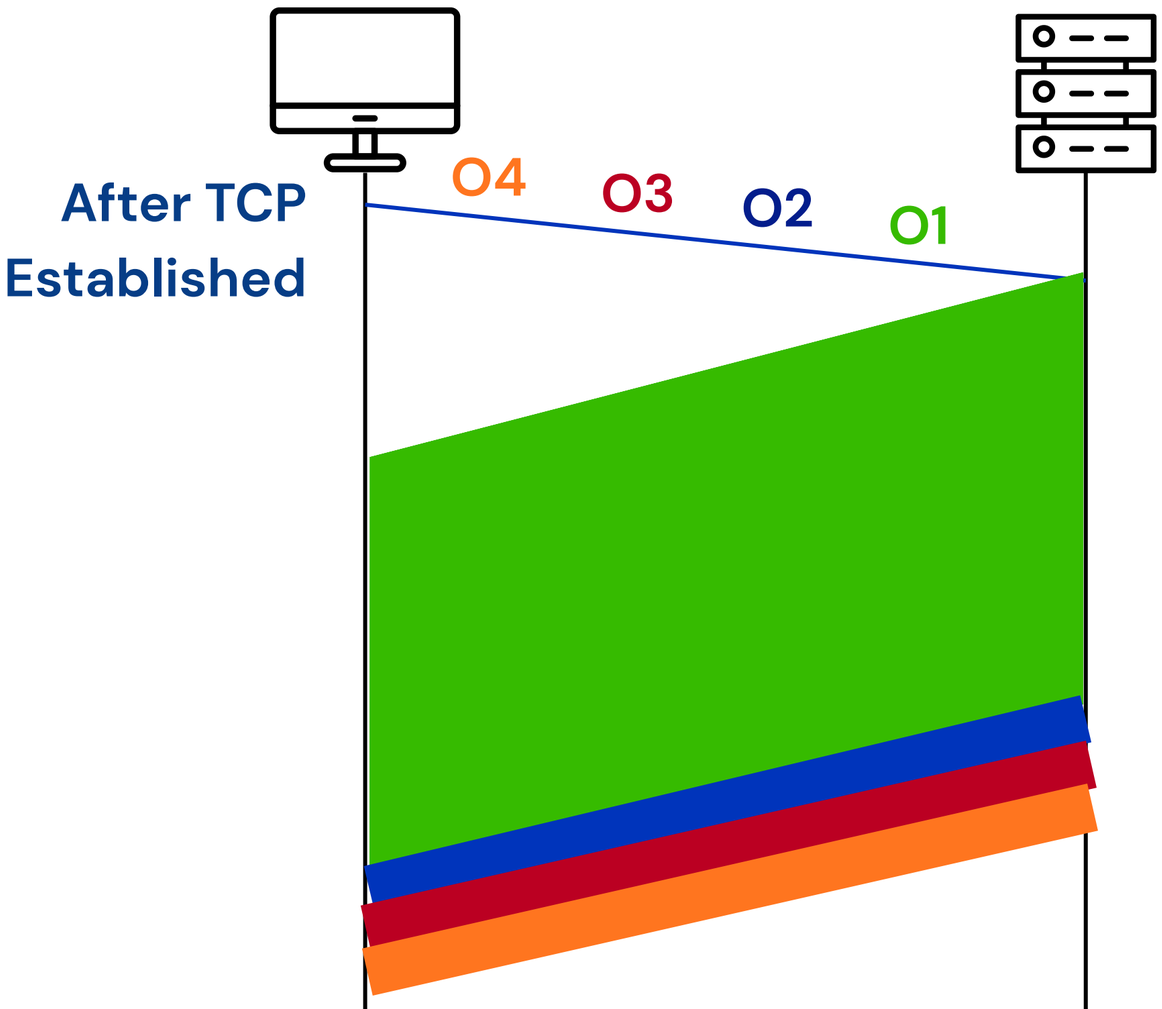

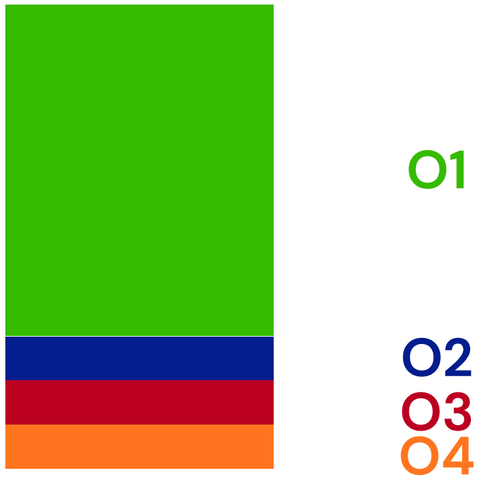

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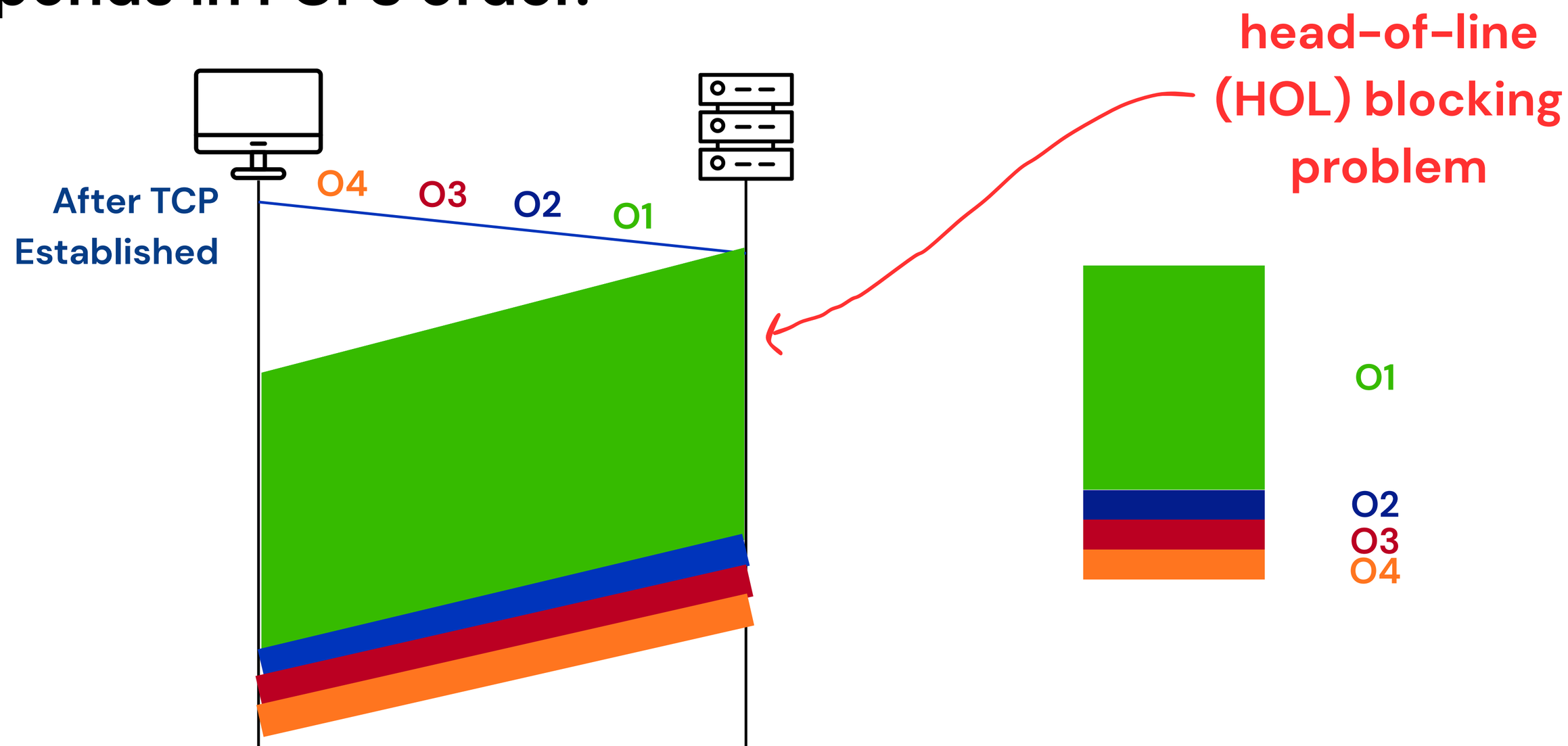

HOL Blocking (HTTP/1.1)

- A slow first request stalls subsequent requests
- If a TCP segment is lost, subsequent segments cannot be delivered to the application until the lost one is retransmitted, stalling the data stream.

HTTP/2

Objects divided into frame. Frame transmissions interleaved.

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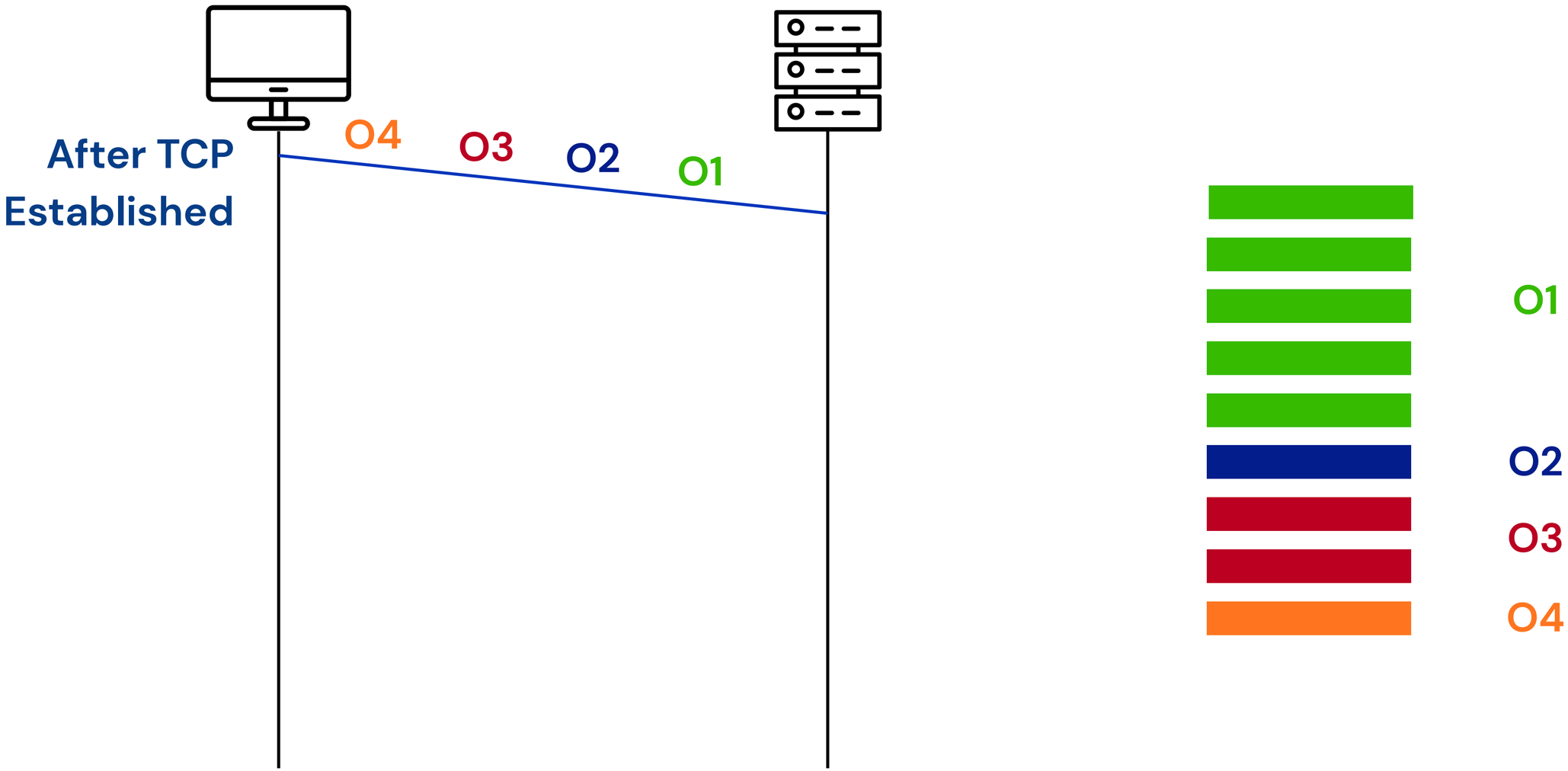

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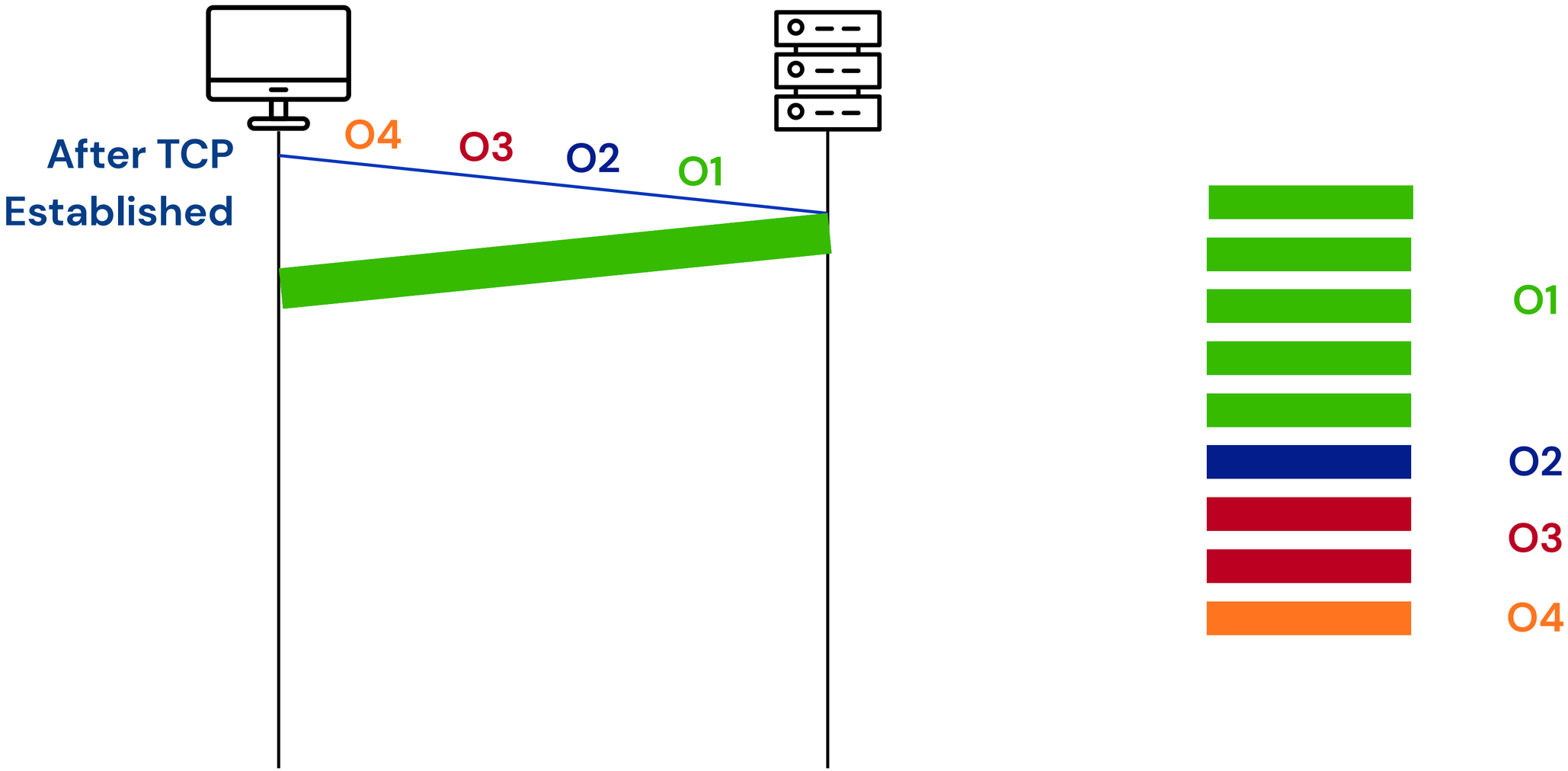

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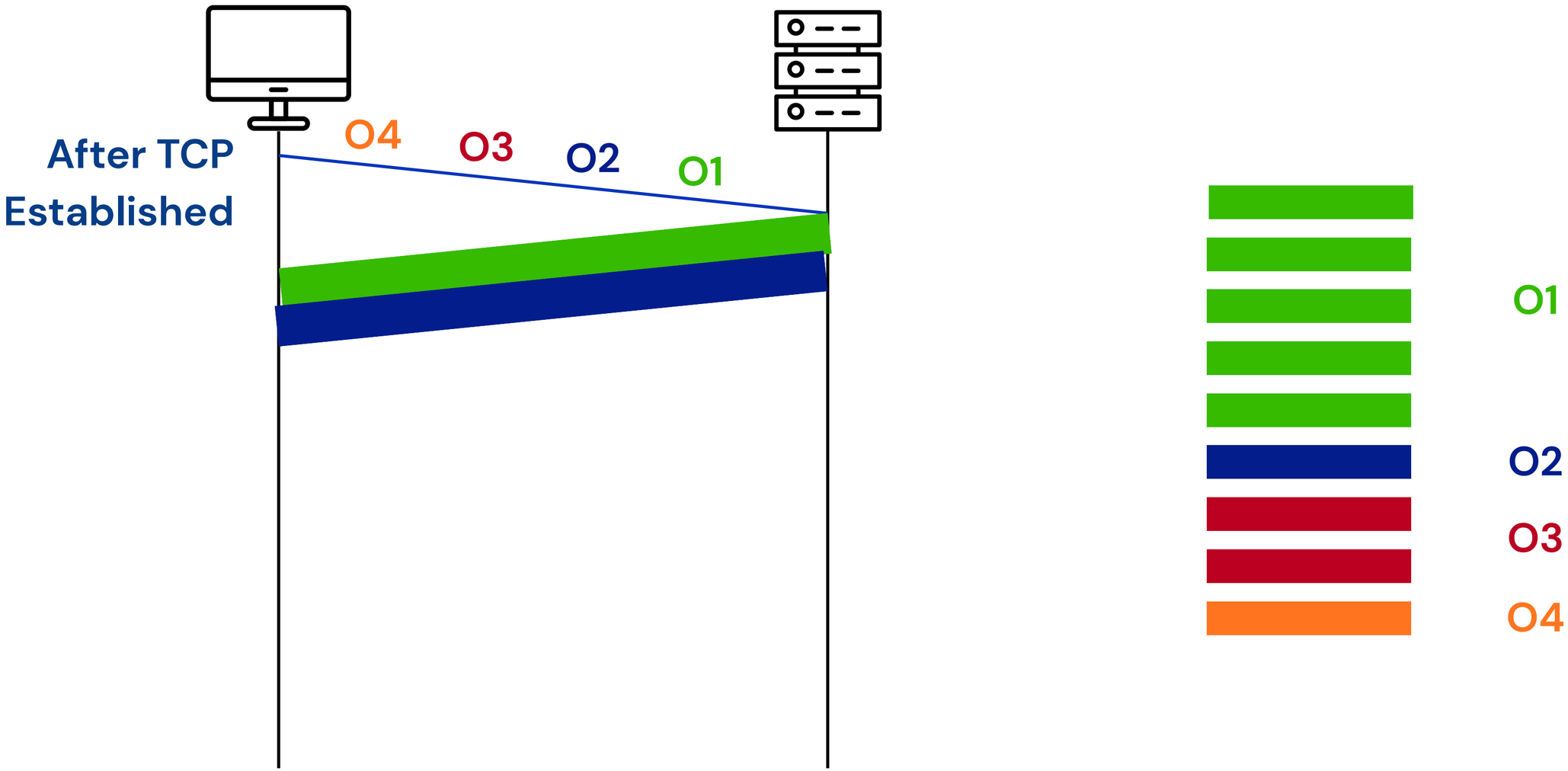

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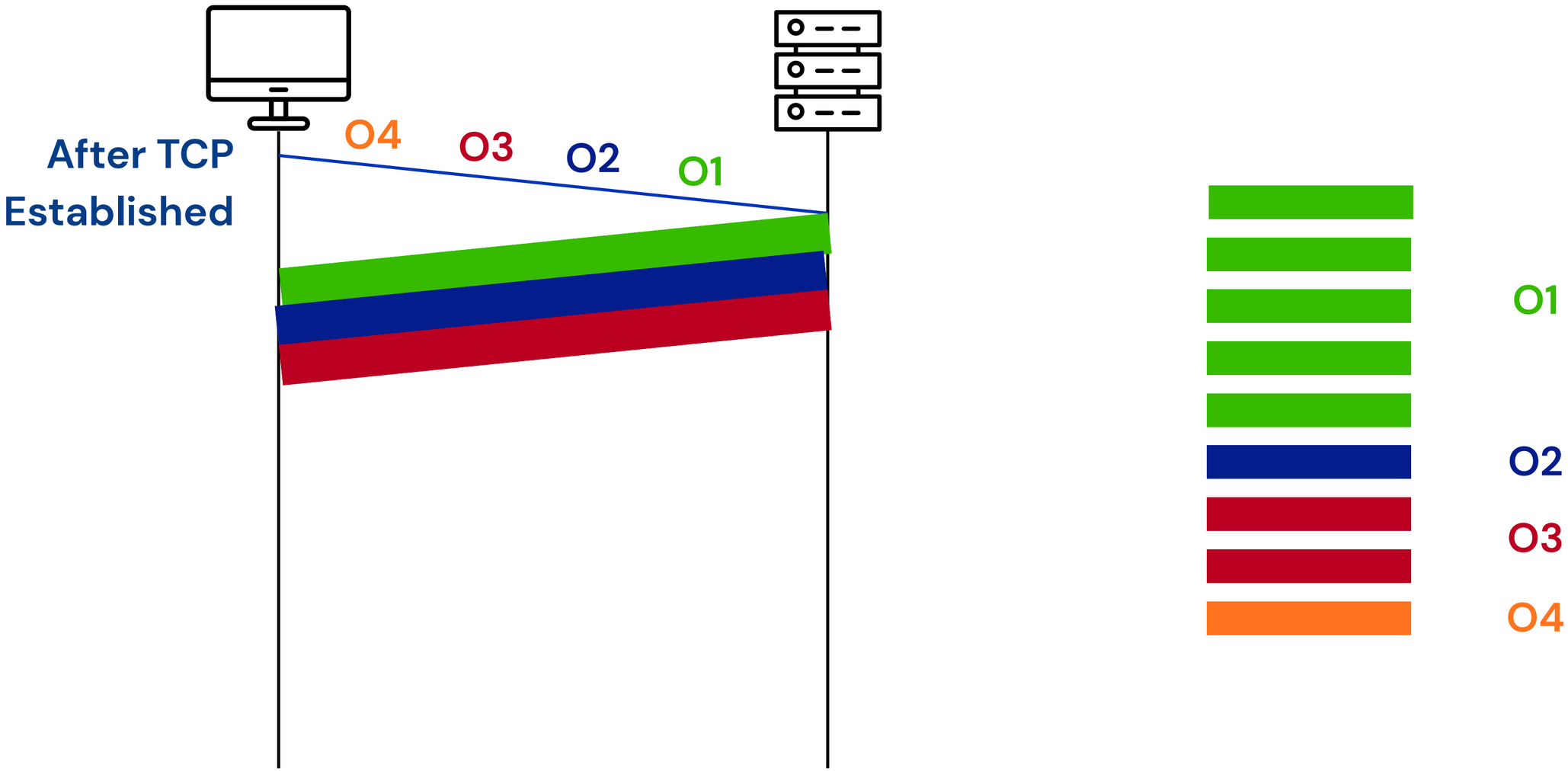

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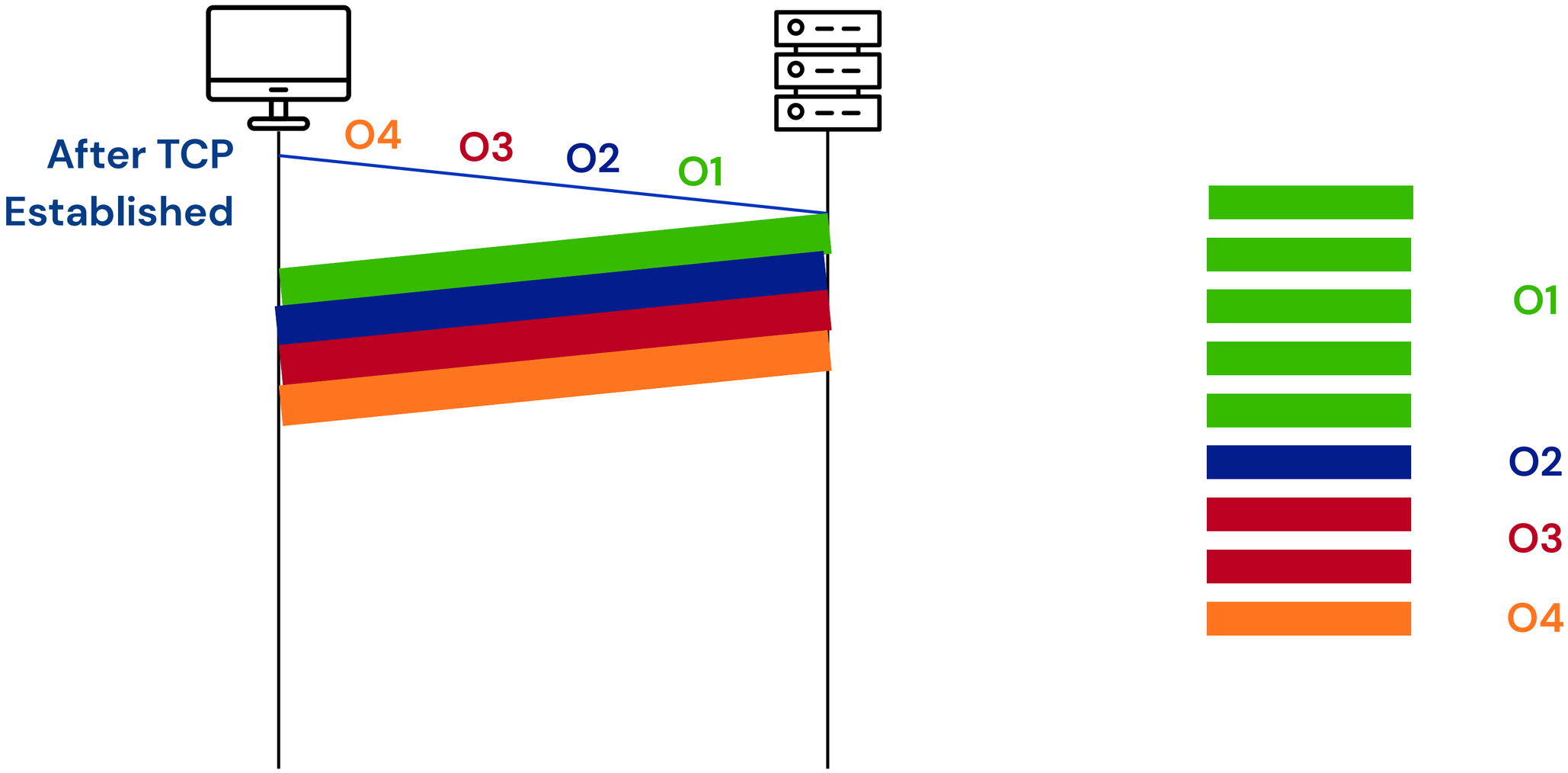

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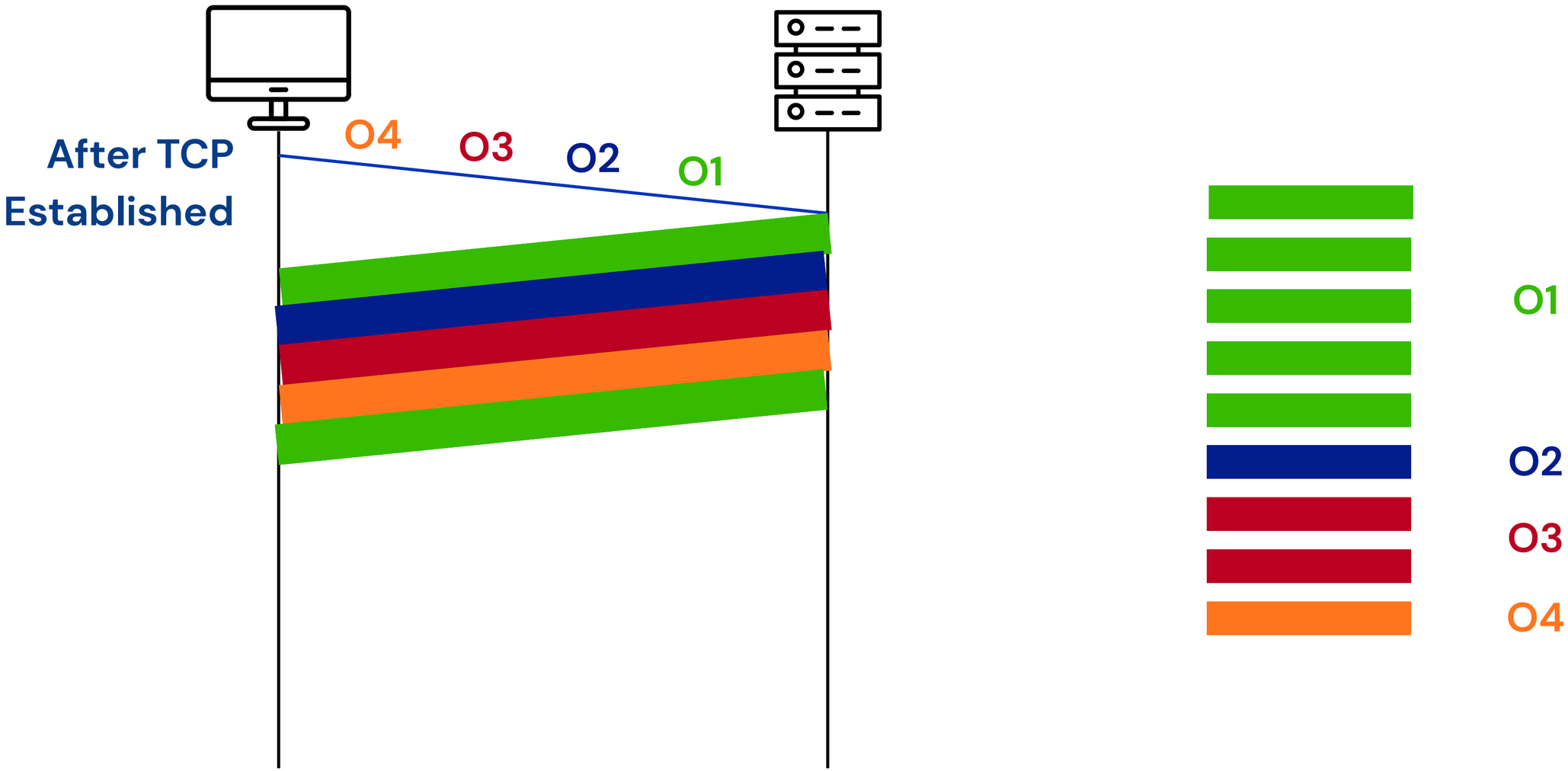

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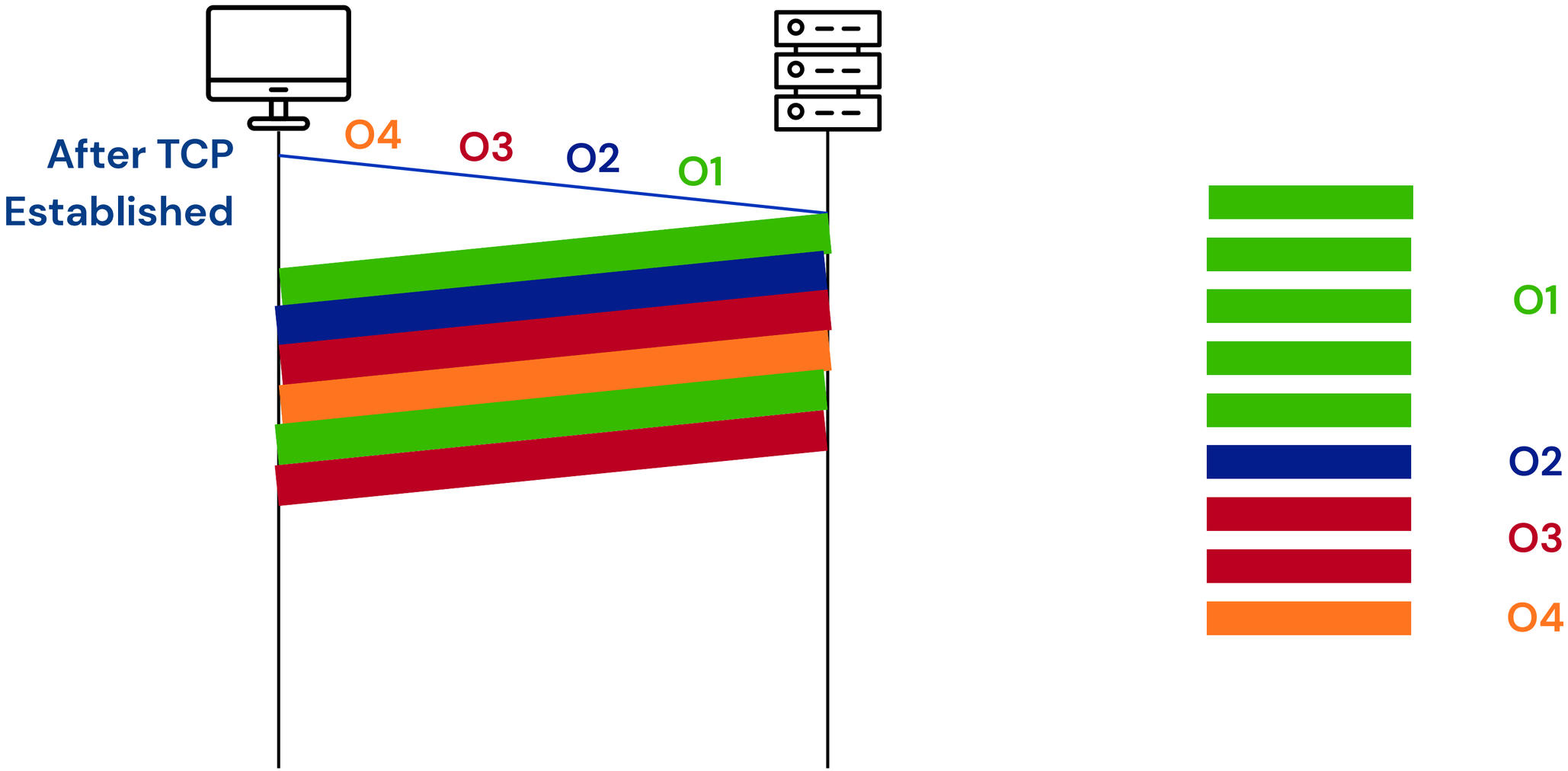

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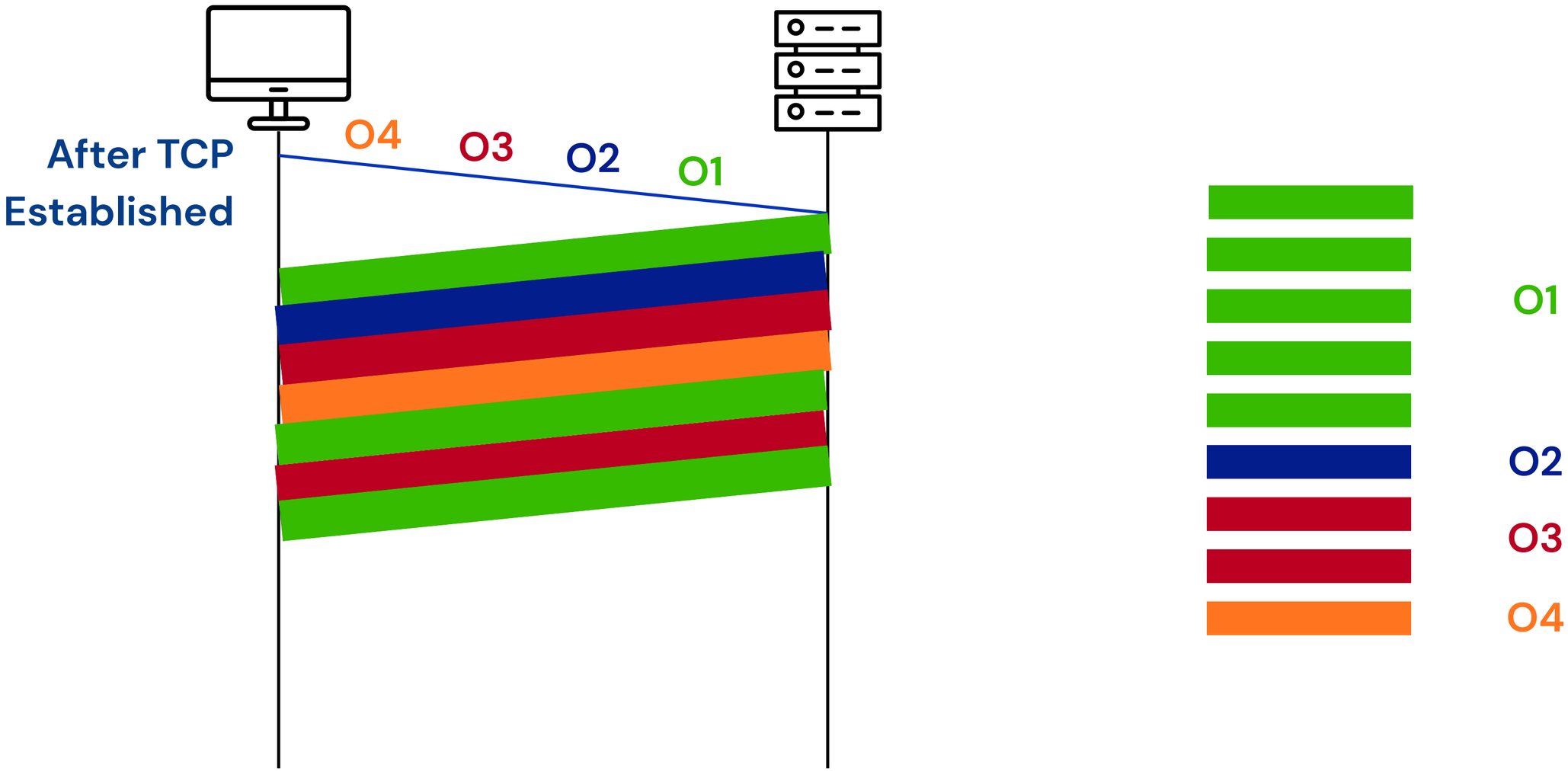

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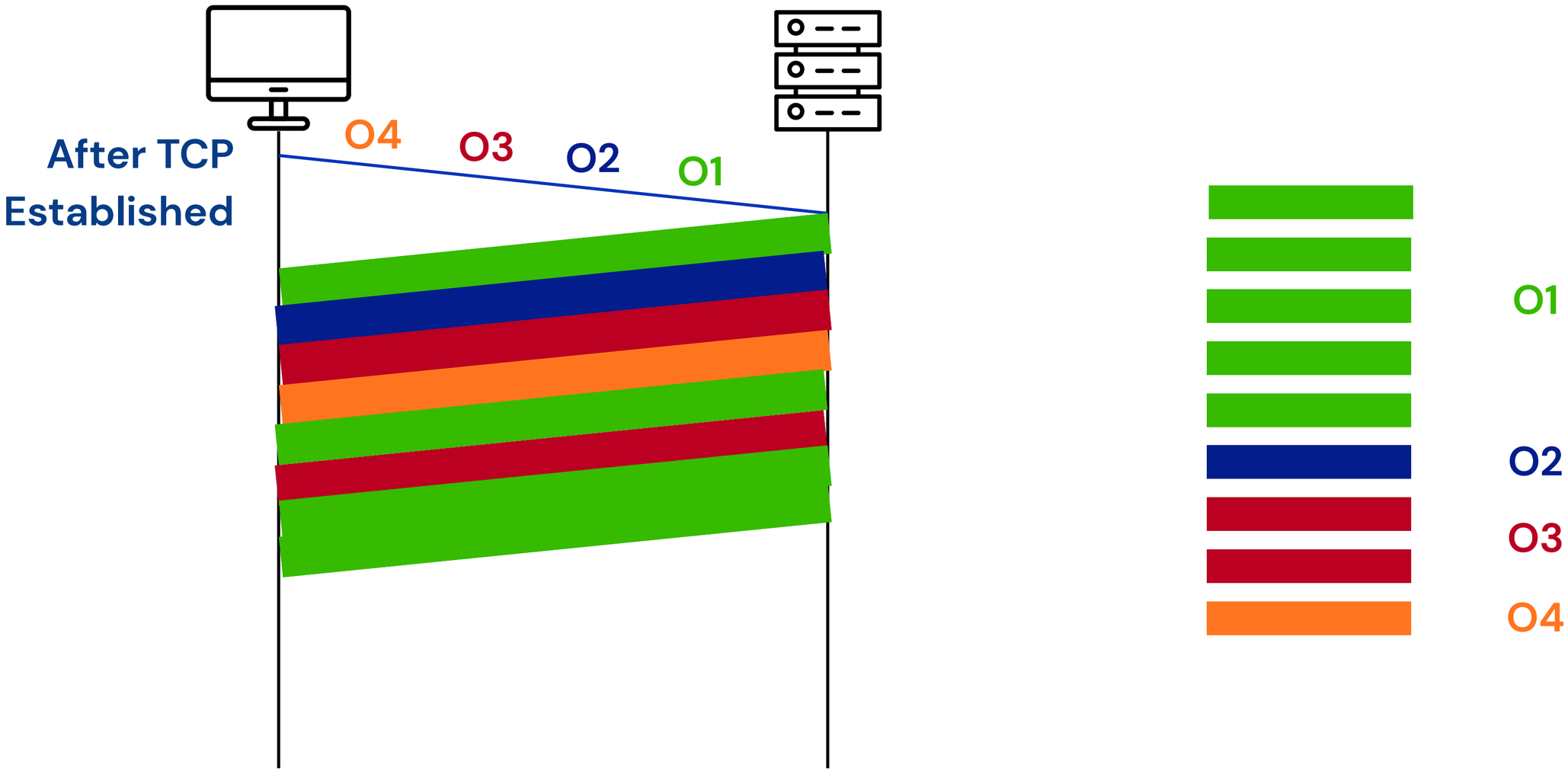

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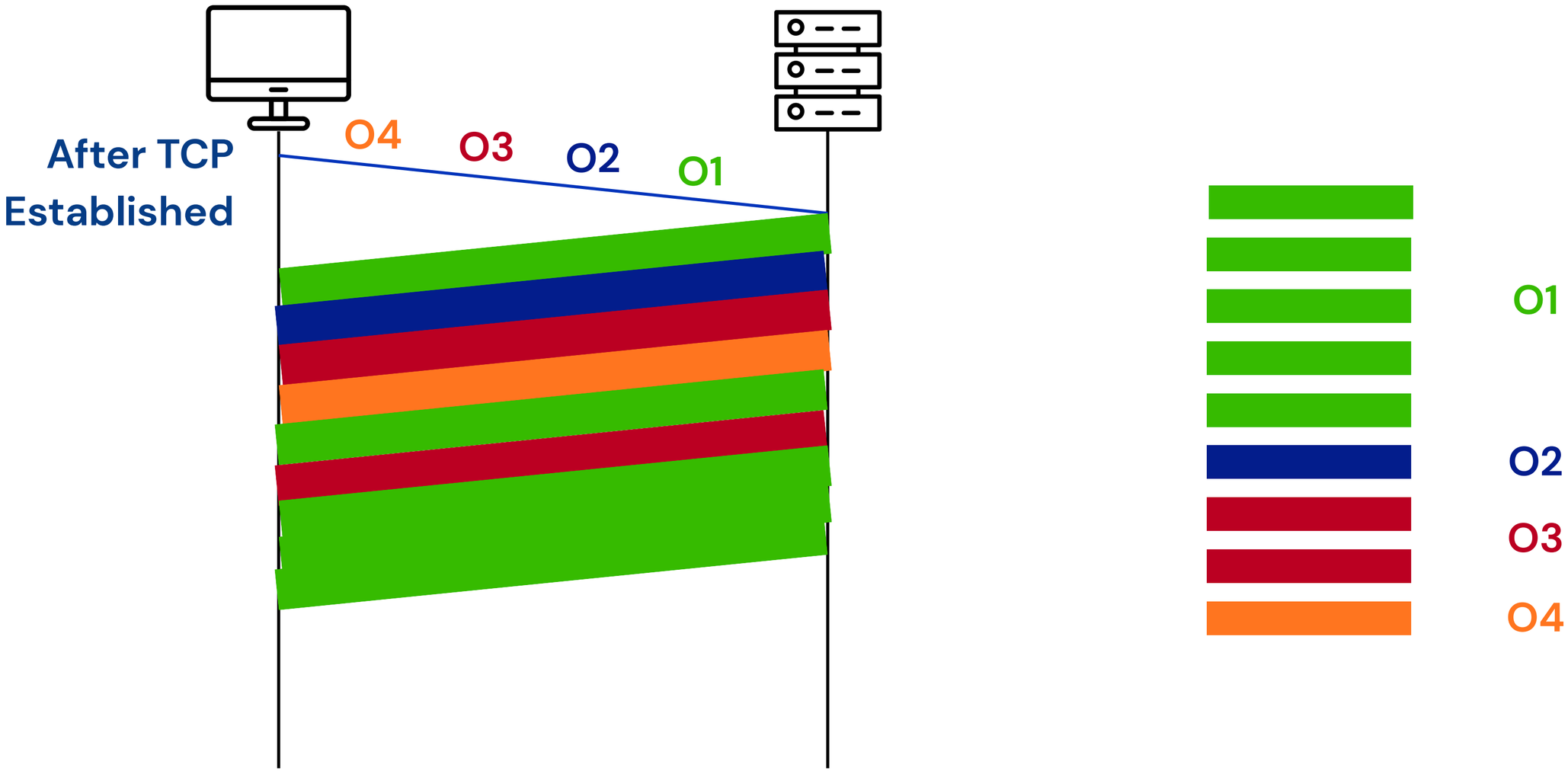

HOL Blocking Remains

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HTTP/3

Replaces TCP with QUIC (which runs over UDP)

- QUIC handles multiplexing independently; a lost packet only affects its specific stream, not the entire connection.
- Faster connection establishment
- Improved congestion control

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What Screams “I’m Insecure”?
Discussion
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1.4k
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1

End of Slide